Aditi Roy Adri

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Summary

Computer Science and Engineering graduate from BRAC University with a strong interest in machine learning. Experienced in training and evaluating models with Python and PyTorch, and in building full-stack web applications.

Education

- **BRAC University, Dhaka** *2023 – 2026*
  Bachelor of Science in Computer Science and Engineering **CGPA:** 3.83/4.00
- **Holy Cross College, Dhaka** *2021*
  Higher Secondary Certificate (HSC) **GPA:** 5.00/5.00
- **Narayanganj High School and College, Narayanganj** *2019*
  Secondary School Certificate (SSC) **GPA:** 5.00/5.00

Technical Skills

- **Programming Languages:** Python, C++, C, Java, PHP, Assembly (Emu8086), Bash
- **Machine Learning & Computer Vision:** Deep Learning, CNNs, Image Segmentation, Object Detection, Transfer Learning, Model Evaluation, YOLO, U-Net
- **Web Technologies:** HTML5, CSS3, JavaScript, React.js
- **Databases:** MySQL, PostgreSQL
- **Libraries & Frameworks:** PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook
- **Graphics & Simulation:** OpenGL (PyOpenGL), Cisco Packet Tracer
- **Tools & Platforms:** Git, GitHub, LaTeX, Linux, VS Code

Projects

- **CARDD – Car Damage Detection with YOLO** *Jun 2026*
  Trained and benchmarked YOLO11-L against YOLOv8-L on the CarDD dataset (4,000 images, 8,740 annotated boxes across 6 damage classes: scratches, dents, cracks, broken lamps, shattered glass, flat tires). YOLO11-L achieved **mAP@50–95 of 0.557**.
  Tech: Python, YOLO11, YOLOv8, PyTorch, Object Detection GitHub
- **Pet Image Segmentation & Breed Classification (U-Net)** *May 2026*
  Deep learning project using U-Net and Attention U-Net architectures for binary pet image segmentation and 37-breed classification on the Oxford-IIIT Pet Dataset. Achieved **∼88% IoU** with a pretrained ResNet-34 encoder. Implemented data augmentation, combined loss functions, and ablation studies.
  Tech: Python, PyTorch, Jupyter Notebook GitHub
- **Humanitarian Classification of Disaster Tweets** *Sep 2026*
  Benchmarked ten models on the HumAID dataset (76,484 tweets, 19 disasters, 10 humanitarian categories with a 59.6:1 class imbalance). **BERT reached 0.768 macro F1**; TF-IDF logistic regression followed at 0.715, outperforming every recurrent model tested.
  Tech: Python, BERT, TF-IDF, Scikit-Learn, NLP GitHub
- **SaveLife – SOS Emergency Response App** *Jul 2026*
  Emergency response platform with one-tap SOS alerts and GPS geolocation, a live responder dashboard, and role-based JWT authentication. React/Vite frontend, Node.js/Express API, PostgreSQL,

deployed on Render.
Tech: React, Vite, Node.js, Express, PostgreSQL, JWT GitHub

- **Roaming Sonic – Travel & Tour Management Platform** *Jan 2026*
  Full-stack MERN web application for Bangladesh tourism featuring tour packages, hotel bookings, bus tickets, guide services, AI chatbot, and referral rewards system. Live-deployed on Vercel.
  Tech: MongoDB, Express.js, React.js, Node.js GitHub | Live Demo

- **Metadata Journal System** *Jan 2026*
  OS-level project implementing file metadata management and journaling mechanisms for maintaining filesystem consistency.
  Tech: C GitHub

Certifications

- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI, Coursera
- **IBM Data Science** (6 courses) — IBM, Coursera: What is Data Science?; Tools for Data Science; Data Science Methodology; Python for Data Science, AI & Development; Python Project for Data Science; Databases and SQL for Data Science with Python
- **Foundations of Cybersecurity** — Google, Coursera
- **BRAC University Programming Contest** — BRAC University Computer Club

Leadership & Activities

- **Women & Child Affairs Secretary**, Executive Committee — Bangladesh Youngster Social Organization (BYSO), Dhaka District Unit, for the 2026–27 session. Leadership role supporting women's and children's welfare initiatives.
- Competitive programming on Codeforces (500+ problems solved in Python).
- Participation in university-level and inter-university programming contests at BRAC University.
- Active volunteer in community and social-welfare activities.
- Vocalist for cultural clubs and events at college and university.

References

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